

The stated unnormalized Mahler bound fails for a cube

Conjecture 00000003552

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Result. With the volume product defined in the source as ordinary volume times polar volume, the symmetric cube in dimension twelve has product

$$\frac{4^{12}}{12!} < \frac{1}{2} \leq \frac{8}{\pi^2}.$$

This contradicts the stated dimension-independent lower bound. In dimension three, the cube's product is $32/3$, which also differs from the claimed extremal value $8/\pi^2$.

An admissible body and its actual polar

Let $n \geq 1$ and use standard coordinates and Lebesgue measure on $\mathbb{R}^n$. Put

$$C_n = [-1, 1]^n, \quad C_n^\circ = \{y \in \mathbb{R}^n : \sum_{i=1}^n x_i y_i \leq 1 \text{ for every } x \in C_n\}.$$

The cube is compact, convex and centrally symmetric, and the origin is an interior point. Thus it is an admissible symmetric convex body. Its polar is exactly

$$C_n^\circ = \{y : \sum_{i=1}^n |y_i| \leq 1\}.$$

Indeed, for $x \in C_n$ we have $\sum_i x_i y_i \leq \sum_i |y_i|$. Conversely, choose $x_i = 1$ when $y_i \geq 0$ and $x_i = -1$ otherwise. This vector lies in C_n and gives $\sum_i x_i y_i = \sum_i |y_i|$. Hence the polar is the genuine cross-polytope determined by the defining pairing.

Lebesgue volumes

The box product formula gives $|C_n| = 2^n$. In each orthant the polar is a coordinate reflection of the standard simplex $\{t_i \geq 0, \sum_i t_i \leq 1\}$, whose volume is $1/n!$. The orthants overlap only on coordinate hyperplanes of measure zero, so

$$|C_n^\circ| = \frac{2^n}{n!}, \quad |C_n| |C_n^\circ| = \frac{4^n}{n!}.$$

The Lean proof evaluates the same Lebesgue volume directly by Mathlib's volume formula for ℓ^p balls, at $p = 1$:

$$\left| \left\{ x : \sum_i |x_i| \leq 1 \right\} \right| = \frac{(2\Gamma(2))^n}{\Gamma(n+1)} = \frac{2^n}{n!}.$$

No polar volume or volume-product identity is assumed.

The numerical contradiction

At $n = 12$, direct integer arithmetic gives $4^{12}/12! < 1/2$. Since $0 < \pi \leq 4$, we have $\pi^2 \leq 16$ and therefore $1/2 \leq 8/\pi^2$.

Proposition 1. *There is a compact, convex, centrally symmetric body with the origin in its interior whose unnormalized Mahler volume is strictly less than $8/\pi^2$.*

Proof. Take C_{12} . The verified polar and volume computations give the strict inequality above. $\square$

For the ordinary three-dimensional cube, the same computation gives

$$|C_3| |C_3^\circ| = \frac{4^3}{3!} = \frac{32}{3} \neq \frac{8}{\pi^2}.$$

Thus if the source’s word “cube” is read specifically in dimension three, it still does not attain the displayed claimed value under the source’s stated volume convention.

Exact scope and normalization

The source explicitly defines Mahler volume as the product of volume and polar volume; no normalization or dimension restriction is supplied. The dimension-twelve example refutes its literal universal lower bound. The dimension-three calculation separately refutes attainment of the displayed value by the spatial cube. It does not assert a lower-bound violation in dimension three.

This does not refute the standard symmetric Mahler conjecture with the dimension-dependent value $4^n/n!$. Nor does it refute a statement about a normalized product. For example, in dimension two the square has unnormalized product 8, whereas dividing by the squared area π^2 of the Euclidean unit disk gives $8/\pi^2$. Such a normalization would change the quantity defined in the source. The asymmetric conjunct is not needed for this disproof.

Formal verification

The Lean project uses finite real coordinate space with its genuine product Lebesgue measure. It proves cube convexity, compactness, symmetry and origin interior; identifies the polar through the complete universal pairing condition and a sign-vector witness; evaluates the two volumes; and derives both dimension-specific conclusions. The concluding theorem assembles the admissibility conditions and strict lower-bound violation.

Run `lake build` in `lean/` using Lean 4.19.0. Mathlib is pinned to

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The final theorem audits use only standard logical axioms. The proof contains no assumed certificates or proof placeholders.